

AI-Powered Behavioral Anomaly Detection

Technical Report

1. Executive Summary

This report details the implementation and evaluation of a hybrid ML architecture for detecting behavioral anomalies and advanced persistent threats.

2. Assumptions

- Synthetic data assumptions: 500 entities, 200K events, 90 days.
- Attack injection rates: predefined specific fractions for brute force, lateral movement, etc.
- Feature engineering choices: sliding windows capturing time, geo-velocity, resource novelty.
- Model architecture decisions: Autoencoder for profiling, BiLSTM for sequences, XGBoost for multi-class classification.

3. System Architecture

The system comprises a data generation pipeline, feature engineering module, a hybrid ML detection engine (AE + LSTM + XGBoost), an explainability layer (SHAP), and an analyst dashboard.

4. Data Schema

10 fields: entity_id, entity_type, timestamp, source_ip, geo_location, resource_accessed, auth_method, session_duration, command_sequence, device_fingerprint

5. Attack Taxonomy

7 Attack Types:

- brute_force
- impossible_travel
- credential_stuffing
- lateral_movement
- device_spoofing
- low_and_slow_exfiltration

- insider_drift

6. Model Details

- Autoencoder: Baseline Profiler
- BiLSTM: Sequence anomaly detection
- XGBoost: Alert classification and attribution

7. Evaluation Metrics

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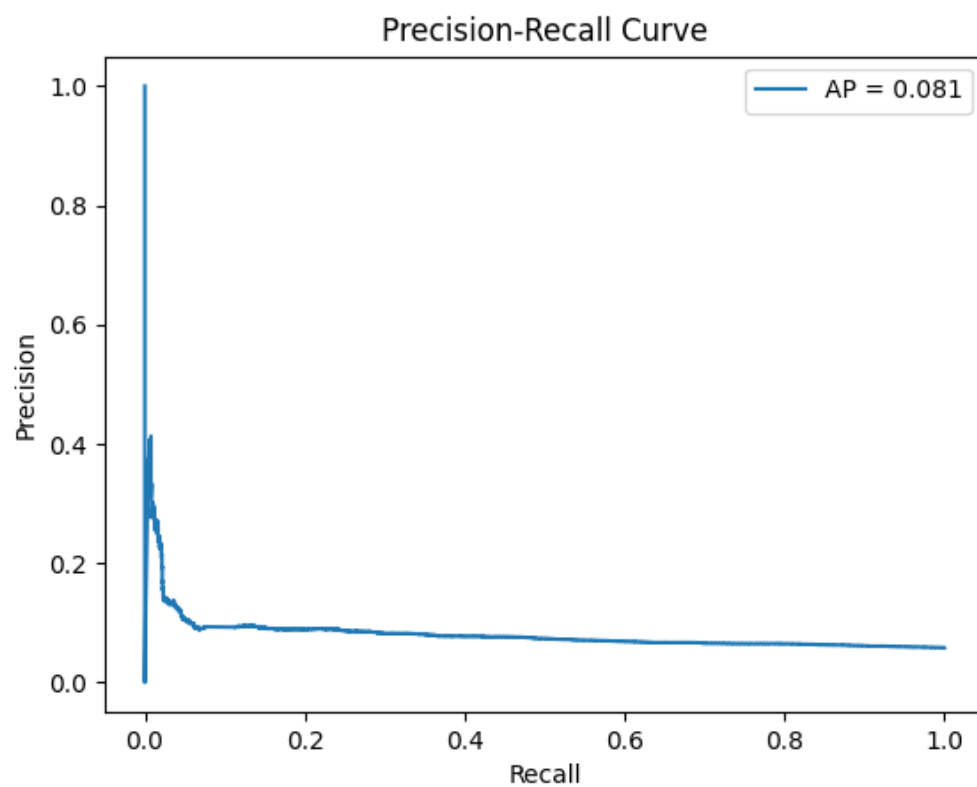
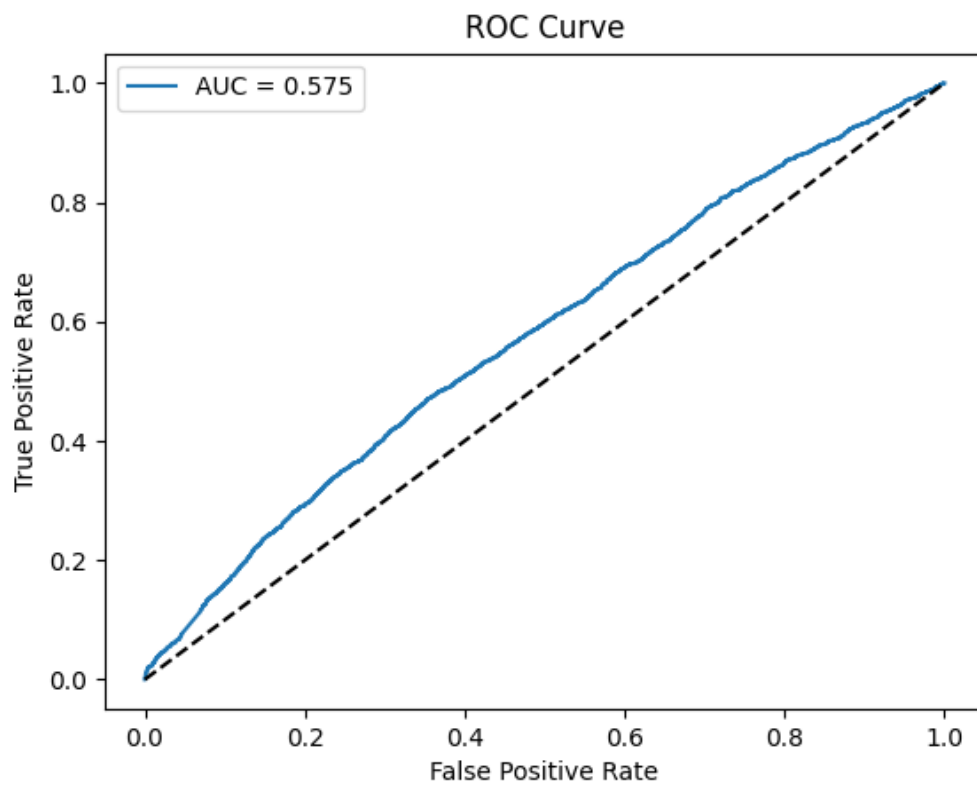
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[Image not found: confusion_matrix.png]

8. False Positive Analysis

FPR at top 1% alert budget: 0.009146798620482831

9. Known Limitations

- Synthetic data may not capture real-world complexity.
- Concept drift handling is window-based, not fully online.
- Cold-start relies on population baseline.
- No graph-based entity-resource modeling.
- Streaming integration requires Kafka/Flink infrastructure.

10. Streaming Feasibility

The architecture can be extended for real-time streaming using Kafka for event ingestion and Apache Flink for stateful feature aggregation (e.g., sliding window counts).